Digital Signal Processing (EC3109)
Fall 2026

Experiment 8

Aim: To verify the first-order difference equation $y(n)=b_1y(n-1)+x(n)$ using both the *filter*() function and iterative calculation for a unit impulse input. Perform the experiment for $b_1 = 2$ and $b_1 = 0.5$ and observe the output.

Equipment required: MATLAB software, Laptop

Algorithm

1. Start
2. Define the discrete time index: n = −5 : 1 : 10.
3. Find the index where n=0.
4. Generate the input sequence x(n) as a unit impulse:
    - x(n) = 1 at n = 0
    - x(n) = 0 otherwise
5. Plot the input sequence using a stem plot.
6. Define first-order IIR filter coefficients:
    - Numerator: b = [1 0]
    - Denominator: a = [1 −2]

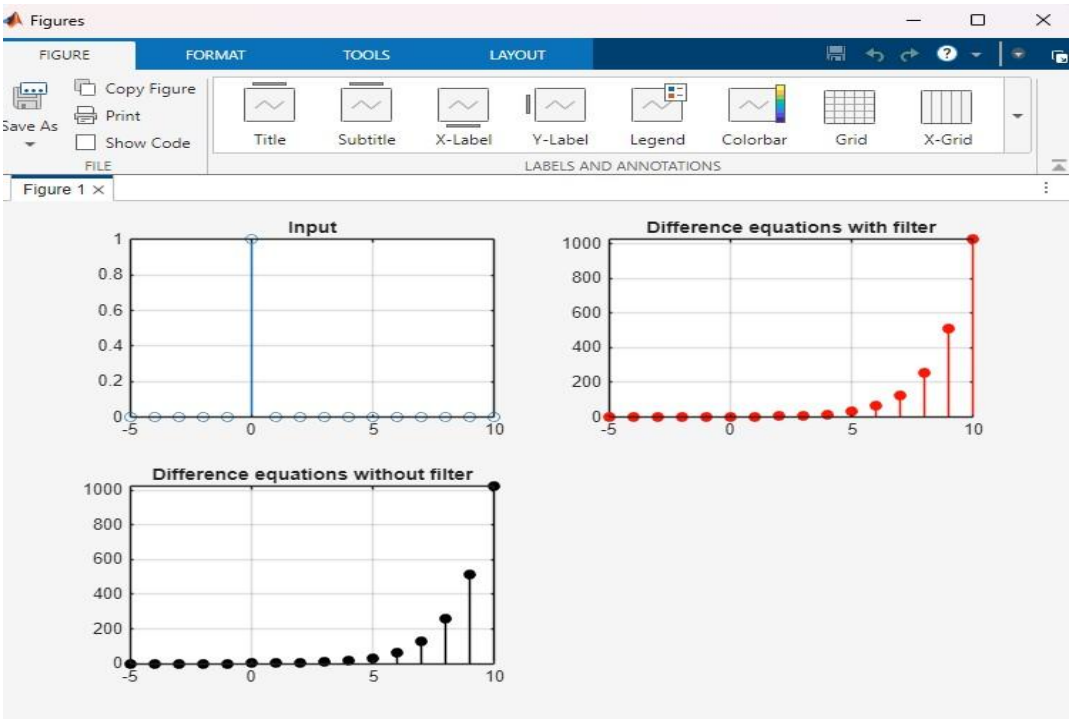
$$y[n] + a_1y[n-1] + a_2y[n-2] + \cdots + a_ny[n-n] = b_0x[n] + b_1x[n-1] + \cdots + b_mx[n-m]$$

7. Compute the output sequence y(n) using the built-in function:
   y(n)=filter(b, a, x(n))
8. Plot y(n) obtained from the filter function.
9. Initialise $y_1(n) = 0$ for all n.
10. For each n:
    - If $n < 0$, keep $y_1(n) = 0$.
    - If $n \geq 0$, compute using the difference equation:

      $$y_1(n)=2y_1(n-1)+x(n)$$
11. Plot $y_1(n)$ obtained from iterative calculation.
12. Compare both y(n) and $y_1(n)$ outputs to verify correctness.
13. End

Results:

1. Output for $b_1 = 2$



2. Output for $b_1 = 0.5$

